Condensation

In empirical work, people look for features in AI systems which correspond robustly to human concepts. In general, these can be hard to find.

One ambitious approach to this problem is to think through from first principles how agents in general might organize information about the world into concepts.

How should information be broken into reusable pieces? Is there one way to do this which many different kinds of agents will converge on?

We will propose two conditions that we think are normative for agents in general: conditions an agent will want its own organization of information to satisfy. Under those conditions, corresponding latent collections in two models determine one another.

Three overlapping observations

Let S, P, Q be independent fair bits.

$$X_1 = (S, P), \qquad X_2 = (S, Q), \qquad X_3 = (P, Q).$$

One latent model assigns

$$Y_{\{1,2\}} = S, \qquad Y_{\{1,3\}} = P, \qquad Y_{\{2,3\}} = Q.$$

Each observation uses two latents. Each latent is recoverable from either observation to which it is assigned.

A latent model

Fix finite-valued observables $(X_i)_{i\in I}$, with $I = \{1, \dots, n\}$, and write $\mathcal{P}^+(I)$ for the nonempty subsets of I.

Each $A \in \mathcal{P}^+(I)$ is an *address*, carrying a latent Y_A assigned to the observations in A. For a family $\mathcal{F} \subseteq \mathcal{P}^+(I)$ of addresses, write $Y_{\mathcal{F}} = (Y_A)_{A\in\mathcal{F}}$ for the corresponding tuple of latents.

The joint law of (X, Y) must satisfy the **latent variable model condition (LVM)**: each observation is determined by the latents assigned to it,

$$H\Big(X_i \mid Y_{\{A\,:\,i\in A\}}\Big) = 0 \qquad (i \in I).$$

Unused slots are constant. This condition does not assert that an agent knows all the realized latent values.

Refresher: entropy and recovery

All variables are finite-valued. Logarithms have base two.

$$H(U) = -\sum_u p(u) \log p(u), \qquad H(U \mid V) = \sum_v p(v) H(U \mid V = v).$$

$$H(U \mid V) = 0 \quad \iff \quad U = f(V) \text{ almost surely, for some } f.$$

The chain rule and conditioning inequality:

$$H(U, V \mid W) = H(U \mid W) + H(V \mid U, W),$$
$$H(U \mid V, W) \leq H(U \mid W).$$

Refresher: dependence

$$I(U;V \mid W) = H(U \mid W) - H(U \mid V,W) \geq 0.$$

This is zero exactly when U and V are conditionally independent given W. Omitting W gives $I(U;V)$.

For laws p, q on the same finite set,

$$D_{\mathrm{KL}}(p\|q) = \sum_u p(u) \log \frac{p(u)}{q(u)} \geq 0.$$

KL divergence appears in the score interpretation. The correspondence proof uses the chain rule and nonnegativity.

LVM and reconstruction

LVM: the assigned latents determine the answer.

$$H\Big(X_i \mid Y_{\{A\,:\,i\in A\}}\Big) = 0.$$

Reconstruction: each contributing observation recovers the slot.

$$H(Y_A \mid X_i) = 0 \qquad \text{whenever } i \in A.$$

An address alone does not imply the second condition.

In our example, $Y_{\{1,2\}} = S$ is recoverable from both (S, P) and (S, Q).

Upward-closed families

A family $\mathcal{F} \subseteq \mathcal{P}^+(I)$ is **upward closed** if

$$A \in \mathcal{F}, \quad A \subseteq B \quad \Longrightarrow \quad B \in \mathcal{F}.$$

Write $\supseteq A = \{C \in \mathcal{P}^+(I) : A \subseteq C\}$ for the addresses containing A; every such family is upward closed. For $I = \{1, 2, 3\}$,

$$\supseteq\{1\} = \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\},$$
$$\supseteq\{2\} = \{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}.$$

Their intersection is $\{\{1,2\}, \{1,2,3\}\}$.

These are families of addresses. $Y_{\mathcal{F}}$ is a tuple of variables.

The Markov condition

For every pair of upward-closed families $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}^+(I)$, require

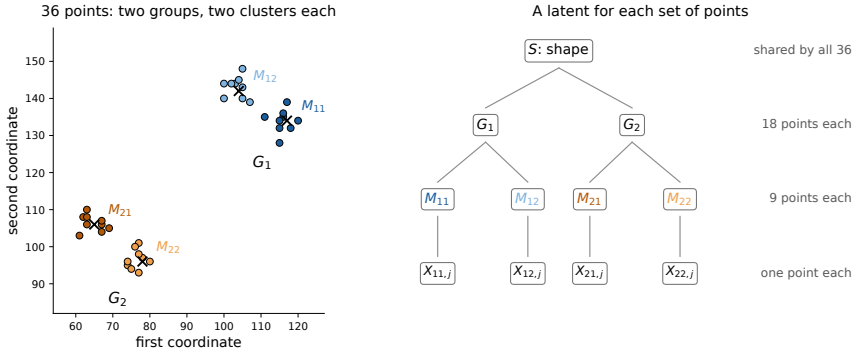
$$I\Big(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F}\cap\mathcal{G}}\Big) = 0.$$

Given the latents at the addresses in the intersection $\mathcal{F} \cap \mathcal{G}$, the two collections are conditionally independent.

For approximate models, record the largest defect:

$$\eta_Y = \max_{\mathcal{F},\mathcal{G}\ \text{upward closed}} I\Big(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F}\cap\mathcal{G}}\Big).$$

A hierarchical model of 36 labelled points



Each point is determined by the latents whose sets contain it: its shape, group centre, cluster centre, and its own position in the cluster.

Why we ask for the conditions only approximately

The cluster center specifies a distribution for its points, not the points themselves, so a point does not determine the center it was drawn from:

$$H(M_{gc} \mid X_i) > 0.$$

Every kernel here has full support: no finite set of points pins a parameter.

LVM and the Markov condition hold exactly here. The reconstruction condition holds only approximately.

So we ask for reconstruction up to a defect: for $i \in A$,

$$H(Y_{\supseteq A} \mid X_i) \leq \delta,$$

and prove a correspondence whose error grows with δ and η.

Two ways the conditions can fail

Return to $X_1 = (S, P), X_2 = (S, Q), X_3 = (P, Q)$ with independent fair bits.

Copy each observation into its singleton slot. LVM and reconstruction both hold, but the common slots are constant:

$$I(Y_{\supseteq\{1\}}; Y_{\supseteq\{2\}} \mid Y_{\supseteq\{1,2\}}) = 1.$$

Put (S, P, Q) only in the top slot. The Markov condition holds, but

$$H(Y_{\{1,2,3\}} \mid X_1) = H(S, P, Q \mid S, P) = 1.$$

The first model hides shared information below constant common slots. The second assigns information to a slot that one observation cannot recover.

Perfect condensation

A latent model is a perfect condensation precisely when

- the Markov condition holds exactly,
- reconstruction holds: $H(Y_A \mid X_i) = 0$ whenever $i \in A$.

LVM is already part of being a latent model.

The combination constrains where information can be placed. Not every observable law admits a perfect condensation.

The collections we will compare

The tuple $Y_{\supseteq A}$ contains the slot at A and all its supersets. For example,

$$Y_{\supseteq\{1,2\}} = (Y_{\{1,2\}}, Y_{\{1,2,3\}}).$$

Our theorem compares $Y_{\supseteq A}$ and $Z_{\supseteq A}$ in two models. Individual slots may contain different amounts of duplicated information.

Note that if we are using the latent at A to predict the observations in A, every latent at a $C \supseteq A$ is assigned to those observations too, so it is already part of what we would need. A family we would actually use to predict an observable is closed upward.